MODULE 6.1

Zone-Based Policy Firewalls

ZPF Overview | Benefits and Design

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Topics:

- 6.1.1 Benefits of a ZPF
- 6.1.2 ZPF Design
- Summary
- References

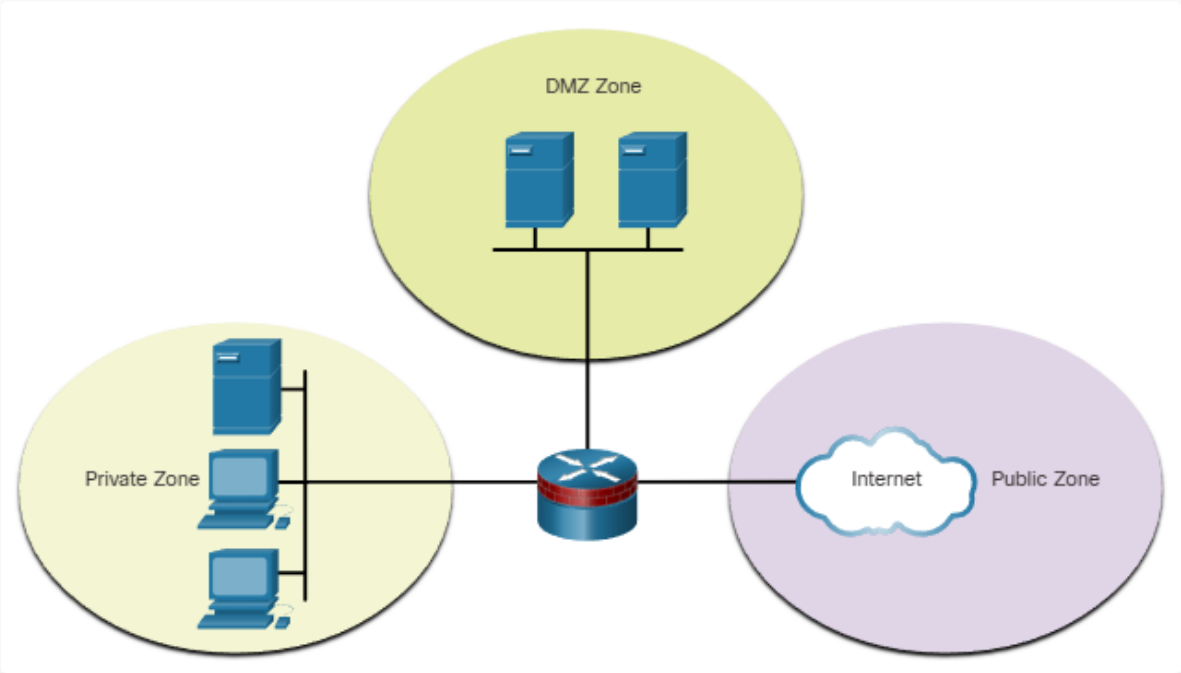
Benefits of a ZPF

- **Two configuration models:** Cisco IOS Firewall supports two ways to build a policy — Classic Firewall, the traditional model where policy is applied directly on interfaces, and Zone-Based Policy Firewall (ZPF), where interfaces are assigned to security zones and policy is applied to traffic moving between zones.

- **Why zones instead of interfaces:** with ZPF, if a new interface is added to a zone, every host connected to it automatically inherits that zone's existing policy toward every other zone — no new interface-specific rules to write.

- **Basic security zone topology:** a simple three-zone network groups interfaces into a Private zone, a DMZ zone, and a Public zone, each attached to the same router.

- **The real benefit:** policy scales with the network. Growing a zone by adding interfaces doesn't mean rewriting the firewall configuration, because the policy lives on the zone, not the wire.



Abbreviations: DMZ = Demilitarized Zone. ZPF = Zone-Based Policy Firewall.

ZPF Design: The Four-Step Process

- **Step 1 — Determine the zones:** decide how the network should be divided, grouping interfaces by the security level and traffic behaviour they actually need.
- **Step 2 — Establish policies between zones:** for every pair of zones that needs to exchange traffic, decide exactly what should be permitted, denied, or inspected.
- **Step 3 — Design the physical infrastructure:** work out what hardware and topology — how many routers or firewalls, and how they connect — is actually needed to realize the zone plan.
- **Step 4 — Identify subsets within zones and merge traffic requirements:** for each firewall device, identify which zone subsets are connected to its interfaces and merge their traffic requirements. Multiple zones might be indirectly attached to a single interface, which results in a device-specific interzone policy.

From the course: *"Implementing zone subsets is beyond the scope of this curriculum" — step 4 matters conceptually, but in practice most designs stop at step 3.*

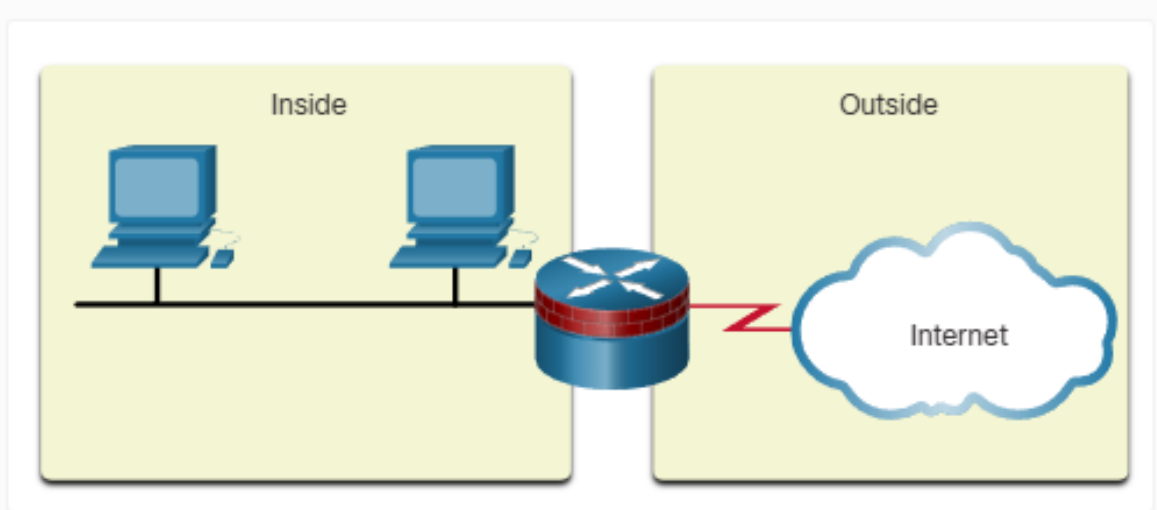
ZPF Design Examples: Starting Simple

- **LAN-to-Internet:** the simplest possible ZPF — just two zones, Inside and Outside, with a deny-by-default policy between them and nothing else to configure.

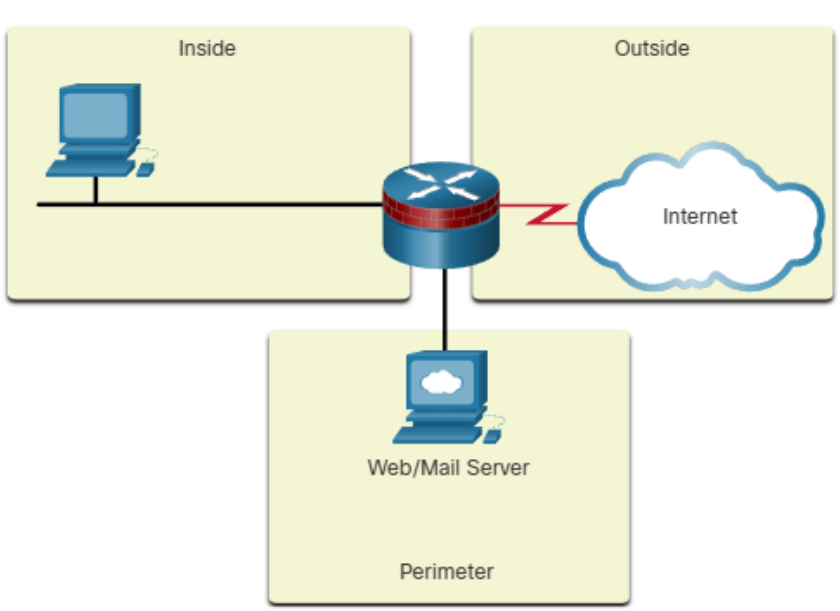
- **Firewall-with-public-servers-1:** adds a third zone, Perimeter, hanging off the same router, hosting a Web/Mail server that needs to be reachable from the Internet without exposing the Inside zone.

- **Same hardware, more zones:** both designs use just one router — the extra zone only costs an extra interface and an extra interzone policy, not extra hardware.

LAN-to-Internet



Firewall-with-public-servers-1



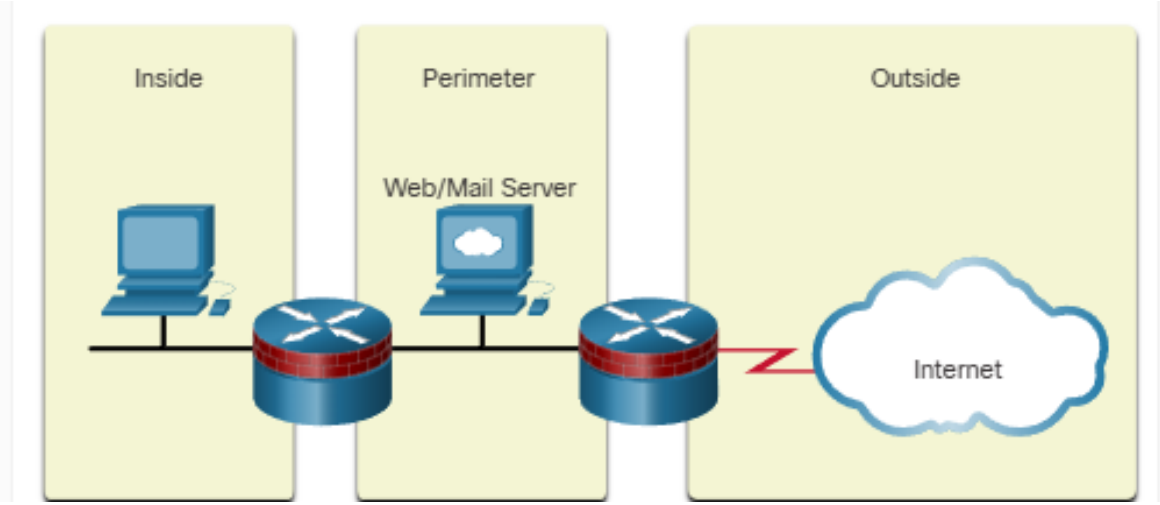
ZPF Design Examples: Splitting and Duplicating

- **Firewall-with-public-servers-2:** splits the single router into two — one firewall between Inside and Perimeter, a second between Perimeter and Outside — so a compromise of the perimeter server doesn't put the inside router's rule set at risk.

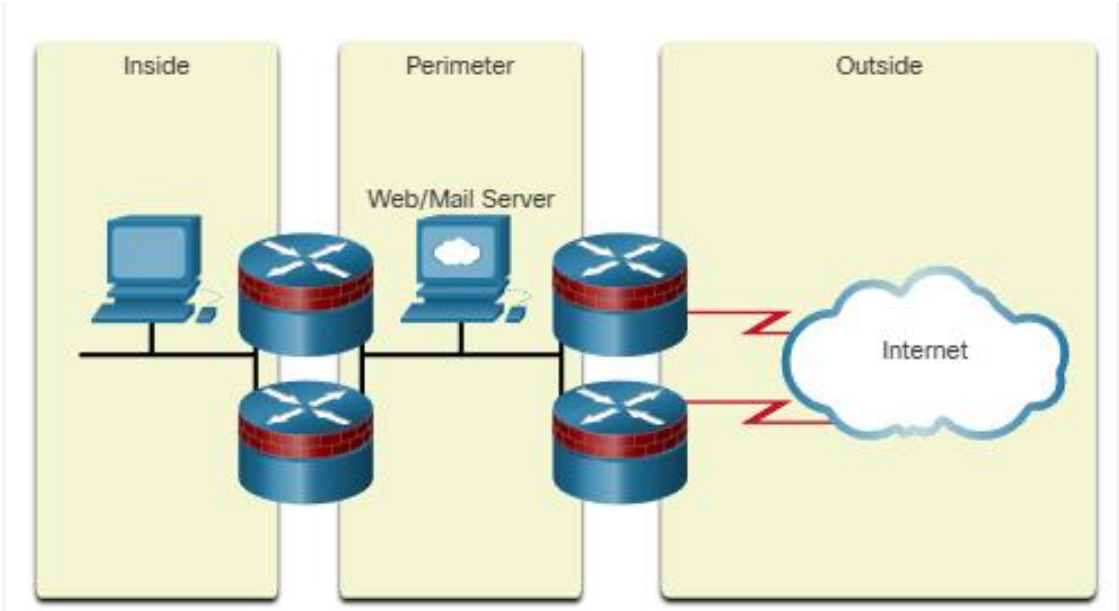
- **Redundant Firewalls:** duplicates each firewall so that if one fails, its pair takes over without dropping the zone's protection — the goal here is availability, not new zones.

- **Trade-off:** more hardware and more to manage, in exchange for isolating failures and limiting the blast radius of a single compromised device.

Firewall-with-public-servers-2



Redundant Firewalls

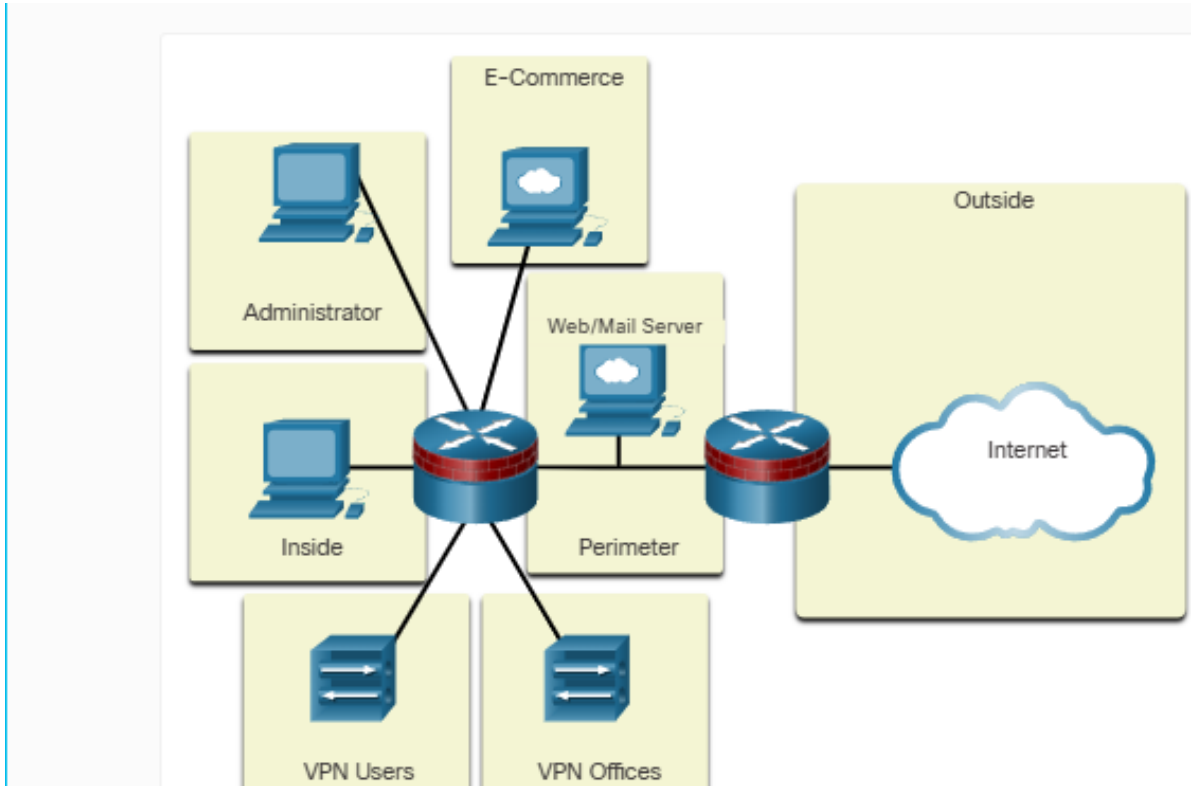


ZPF Design Examples: Complex Firewall

- **Six zones at once:** Administrator, Inside, VPN Users, VPN Offices, E-Commerce, and Perimeter (Web/Mail Server) all meet at one firewall, with Outside reached through a second.

- **Every spoke is its own zone:** VPN Users and VPN Offices are kept separate from the regular Inside zone, so remote-access traffic doesn't automatically inherit the same trust as a wired LAN host.

- **This is the ceiling, not the floor:** real deployments rarely need six zones — but the same four-step design process scales from the two-zone LAN-to-Internet example all the way up to this.



Summary

- **6.1.1 —** ZPF replaces per-interface rules with per-zone policy: put interfaces in a zone once, and every host on that zone inherits the same rule set, including hosts added later.

- **6.1.2 —** every ZPF design follows the same four steps: determine the zones, establish policies between them, design the physical infrastructure, then merge traffic requirements for any interface that touches more than one zone.

- **The five examples** are really one idea repeated at increasing scale: start from Inside/Outside, add a perimeter zone for public servers, split that into two firewalls, add redundancy, then add as many special-purpose zones (VPN, e-commerce, admin) as the network actually needs.

References

Cisco Networking Academy — Network Defense, Module 6.1: Zone-Based Policy Firewalls

https://www.netacad.com

Cisco — Zone-Based Policy Firewall Design and Application Guide

https://www.cisco.com/c/en/us/support/docs/security/ios-firewall/98628-zone-design-guide.html

Cisco — Zone-Based Policy Firewalls vs. Classic Firewall

https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/sec_data_zbf/configuration/xe-3s/sec-zone-pol-fw-xe-3s-book.html